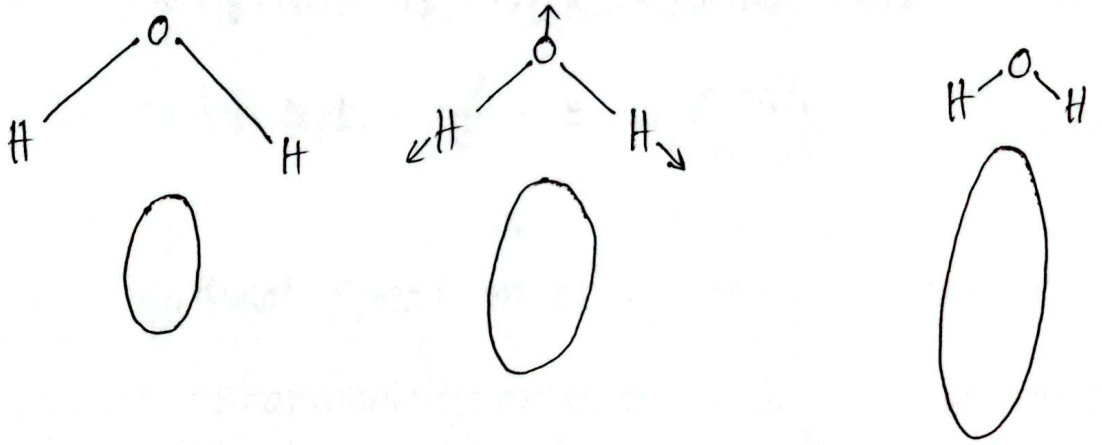
Chapter-4

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(*) 2022, 2016
Explain the Raman activity of the fundamental mode of H_2O?

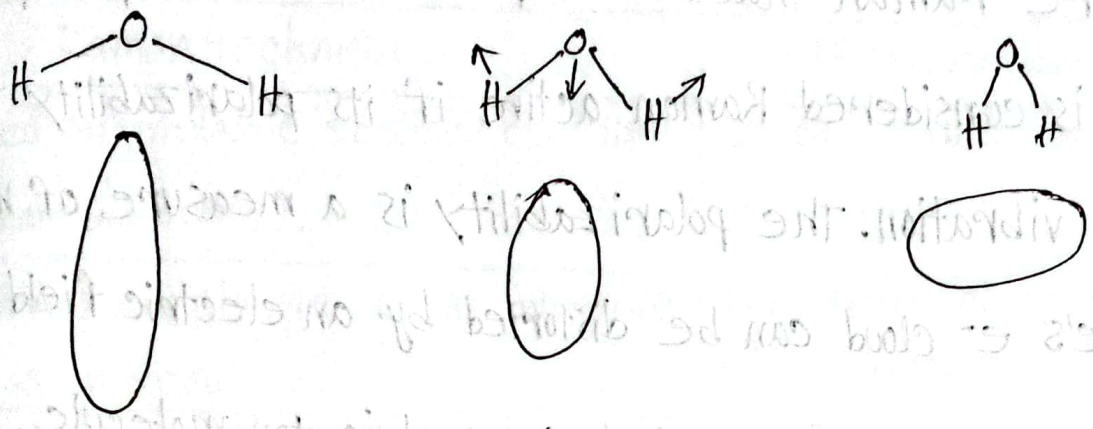
⇒ Fundamental mode of H_2O are $= 3N-6$
$= 3\times3-6 = 3$

3 fundamental modes of H_2O are symmetric, asymmetric stretching & Bending.

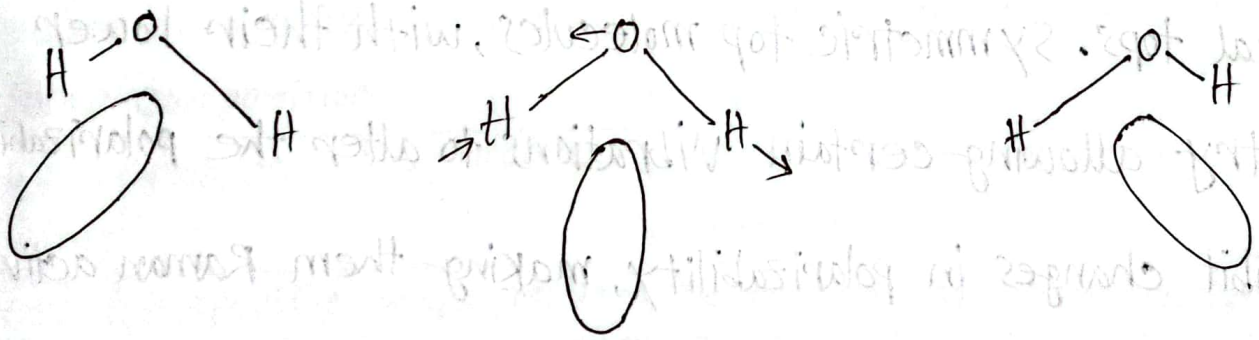


this is symmetric stretching mode of H_2O. During ~~stretching~~ symmetric stretching the molecule increases & decreases in size. when a bond is stretched, the e^- forming it are less firmly held by the nuclei & so the bond becomes more polarizable. Thus the polarizability ellipsoid of H_2O may be expected to decrease in size while the bond stretch, & to increase while they compress but to maintain an approximately constant shape. As there is change in ~~magnitude~~ size of polarizability ellipsoid, so symmetric

stretching mode of H_2O is Raman active



This is bending mode of H_2O. In bending motion the shape of the ellipsoid changes most. For change in shape of polarizability ellipsoid bending mode of H_2O is Raman active.



This is asymmetric mode of H_2O. In this mode both the size & shape remains approximately constant but the direction of the major axis changes markedly. For change in direction this mode is Raman active.

[Change in size/shape/direction = Raman active]
N.B.

(2022) Symmetric top molecules are Raman active but spherical top molecules are raman inactive - explain with example?

⇒ A molecule is considered Raman active if its polarizability changes during vibration. The polarizability is a measure of how easily a molecule's e^- cloud can be distorted by an electric field.

The polarizability ellipsoid for typical symmetric top molecule, plainly rotation about the top axis produces no change in the polarizability but end-over-end rotation will produce a change. Symmetric top molecules exhibit lower symmetry than spherical tops. Symmetric top molecules, with their lower symmetry allowing certain vibrations to alter the polarizability & exhibit changes in polarizability, making them Raman active. such as $CHCl_3$.

The polarizability ellipsoid for spherical top molecule is a spherical surface, & it is evident that rotation of this ellipsoid will produce no change in polarizability. So the pure rotations of spherical top molecules are completely inactive in the Raman. Ex. CH_4, SiH_4.

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2021
⊛ State difference btwn Raman technique & IR vibrational measurement

⇒

Raman technique	IR vibrational measurement
Based on inelastic scattering of light (Raman scattering).	Based on absorption of IR radiation.
Depends on changes in the polarizability of a molecule	Depends on changes in the dipole moment of a molecule.
Detects symmetric vibrations effectively	Detects asymmetric vibrations effictively
Requires a monochromatic light source, such as a laser	Uses a broadband IR radiation source
Vibrations are Raman-active if they cause a change in polarizability.	Vibrations are IR-active If they cause a change in dipole moment.
less sensitive to water interference	Higly sensitive to water interference.
Used in structural analysis, identifying molecular compositions.	Used in functional group analysis, identifying impurities.
Provides informations on symmetric vibrations	Provides information on asymmetric vibrations.
No need for specific sample preparation	Requires specific sample preparation.
can use water as a solvent	cannot use water as a solvent
Higly expensive method	comperatively inexpensive
Raman intensity depend on polarizability of the sample.	Doesnot depend on the sample's polarizability

2021
⊛ Define molecular polarizability?

⇒ M. polarizability is a measure of how easily the e⁻ cloud of an atom or molecule can be distorted by an external electric field. when a molecule is put into a static electric field it

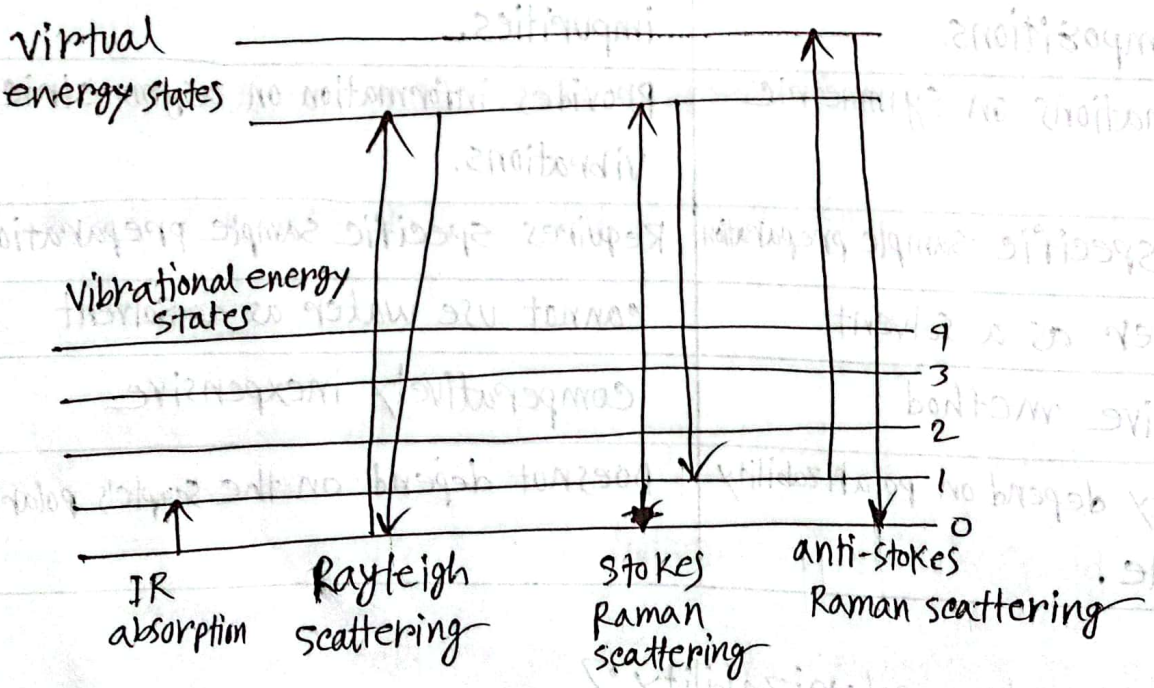
Suffers some distortion, the positively charged nuclei being attracted towards the negative pole of the field, the e^- to the positive pole. This seperation of charge centres causes an induced electric dipole moment to be set up in the molecule & the molecule is said to be polarized. The size of the induced dipole μ, depends both on the magnitude of the applied field, E and on the ease with which the molecule can be distorted. We can write

$$\mu = \alpha E$$

where, α is the polarizability of the molecule.

⊛ 2020 Illustrate quantum theory of Raman spectra?

⇒



⊛ 2020 Write energy expression & selection rule of Raman spectra in case of symmetric top molecules?

energy expression,

$$\varepsilon_{J,K} = BJ(J+1) + (A-B)K^2 \ cm^{-1}, \quad (J=0,1,2,\ldots;\ K=\pm J, \pm(J-1), \ldots)$$

Selection rule for Raman spectra are

$$\Delta K = 0$$

$$\Delta J = 0, \pm 1, \pm 2 \quad (\text{except for } K=0 \text{ states when } \Delta J = \pm 2 \text{ only})$$

where, K = rotational quantum number for axial rotation.

2020

Show that R branch lines appear at $4B, 6B, 8B, 10B \ldots cm^{-1}$ from the exciting line?

⇒ A symmetric top molecules has anisotropic polarizability. This selection rule hold for any K. There are two cases:

① $\Delta J = +1$ (R branch)

$$\Delta E_R = \varepsilon_{J'=J+1} - \varepsilon_{J''=J}$$

$$= 2B(J+1) \ cm^{-1} \quad (J=1,2,3,\ldots \text{ (but } J \neq 0))$$

② $\Delta J = +2$ (S branch)

$$\Delta E_S = \varepsilon_{J'=J+2} - \varepsilon_{J''=J}$$

$$= B(4J+6) \ cm^{-1} \quad (J=0,1,2,\ldots)$$

∴ Allowed R branch lines,

when, $J=1$, $\Delta E_R = 2B(1+1) = 4B \ cm^{-1}$.

When, $\bar{J}=2$, $\Delta\varepsilon_R = 2B(2+1) = 6B$ cm^{-1}

when, $\bar{J}=3$, $\Delta\varepsilon_R = 2B(3+1) = 8B$ cm^{-1}

when, $J=4$, $\Delta\varepsilon_R = 2B(4+1) = 10B$ cm^{-1}

So, R branch lines appear at 4B, 6B, 8B, 10B..... cm^{-1} from the exciting line.

2016

The wavenumber of the incident Radiation in a Raman Spectrometer is 20487 cm^{-1}. What is the wavenumber of the scattered stokes radiation for the $\bar{J}=2 \leftarrow 0$ transition of $^{14}N_2$? [Given $B = 1.9987\ cm^{-1}$] ?

⇒ transition $\bar{J}=0 \leftarrow 0$ so $\Delta\bar{J} = +2$ that mean it is S branch.

Given, $B = 1.9987 cm^{-1}$, $\bar{\nu}_{ex.} = 20487\ cm^{-1}$, $\bar{J}=0$

For S branch, $\bar{\nu}_S = \bar{\nu}_{ex.} - \Delta\varepsilon_S = \bar{\nu}_{ex.} - B(4\bar{J}+6)$ cm^{-1}

$= 20487 - (1.9987 \times 6) = 20475\ cm^{-1}$ Ans.

2015, 2017

If the bond length of H_2 is 0.07417 nm. What would be the spacing of the lines in the spectrum?

⇒ $\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{1.0078 \times 1.0078}{1.0078 + 1.0078} \times 1.6605 \times 10^{-27} = 8.42 \times 10^{-28}\ kg$

$r = 0.07417 \times 10^{-9}\ m$, $c = 3 \times 10^8\ ms^{-1}$

$$\therefore B = \frac{h}{8\pi^2 I c} = \frac{h}{8\pi^2 \mu r^2 c} = 60.5\ cm^{-1}$$

$\therefore$ Line spacing $= 4B = 243.16\ cm^{-1}$ Ans.

2016

(A) Consider the symmetric top molecule $CHCl_3$:

(I) write energy expression?

(II) write selection rule for raman spectra in this case?

(III) write eqn of $\bar{\nu}_R$ & $\bar{\nu}_S$ mentioning their J & ΔJ?

(IV) sketch S branch series only in a figure & show S branch occur at $6B, 10B, 14B, \ldots\ cm^{-1}$ from the exciting line?

$\Rightarrow$ (I) energy, $\varepsilon_{J,K} = BJ(J+1) + (A-B)K^2\ cm^{-1}$ $(J = 0, 1, 2, \ldots;\ K = \pm J, \pm(J-1), \ldots)$

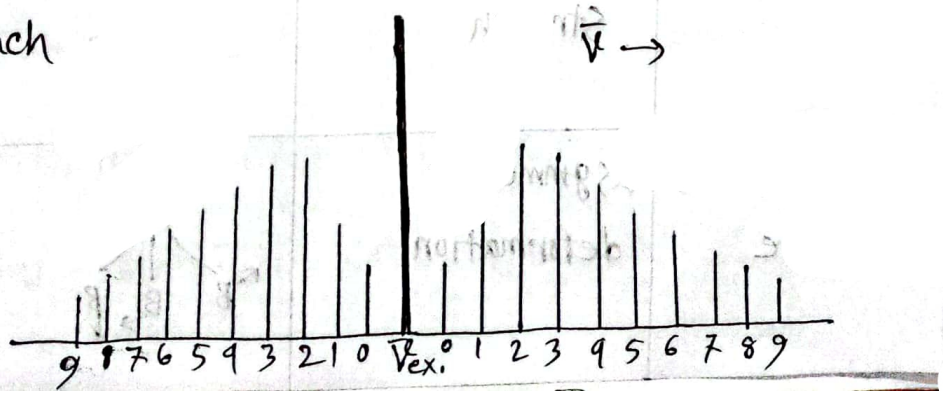
(II) selection rule, $\Delta K = 0$

$\Delta J = 0, \pm 1, \pm 2$ (except for $K=0$ states when $\Delta J = \pm 2$ only)

(III) $\bar{\nu}_R = \bar{\nu}_{ex.} \pm \Delta\varepsilon_R = \bar{\nu}_{ex.} \pm 2B(J+1)\ cm^{-1}$ $(J = 1, 2, \ldots)$

$\bar{\nu}_S = \bar{\nu}_{ex.} \pm \Delta\varepsilon_S = \bar{\nu}_{ex.} \pm B(4J+6)\ cm^{-1}$ $(J = 0, 1, 2, \ldots)$

(IV) Fig. of S branch



If we consider $\bar{\nu}_S = \bar{\nu}_{ex.} \pm B(4J+6)$ cm^{-1} eqn for S branch then (excepting $\bar{\nu}_{ex.}$), taking $J = 0, 1, 2$ gradually, $\Delta\bar{\nu}_S = B(4B+6)$

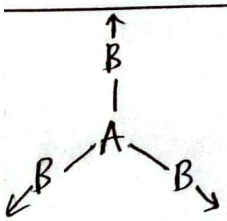
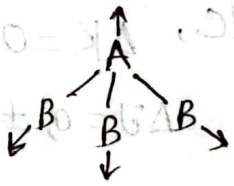
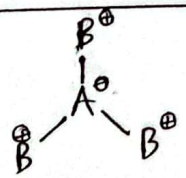
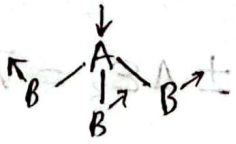
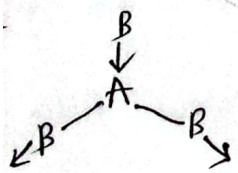
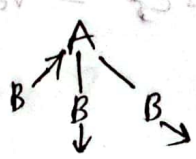
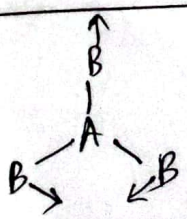
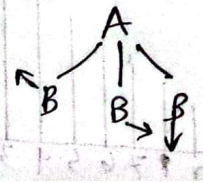
when, $J=0$, $\Delta\bar{\nu}_S = B(4\times0+6) = 6B$ cm^{-1}

when, $J=1$, $\Delta\bar{\nu}_S = B(4\times1+6) = 10B$ cm^{-1}

when, $J=2$, $\Delta\bar{\nu}_S = B(4\times2+6) = 14B$ cm^{-1}

So S branch occur at 6B, 10B, 14B, ... cm^{-1} from existing line.

(~~X~~) 2016 which of the ~~following~~ vibrations of AB_3 are IR or Raman active when it is (i) trigonal planar (symmetric planar) (ii) trigonal pyramidal (symmetric pyramidal)?

symmetric planar	activity (R = Raman, I = Infrared)	vibration	symmetric pyramidal	activity (Raman = R, Infrared = I)
	R = active (pol.) strong I = inactive	Symmetric stretch		R = active (pol.) strong I = active ‖
	R = inactive I = active ‖	out-of-plane symmetric deformation		R = active (pol.) medium I = active ‖
	R = active (depol.) weak I = active ⊥	asymmetric stretch		R = active (depol.) weak I = active ⊥
	R = active (depol.) weak I = active	asymmetric deformation		R = active (depol.) weak I = active ⊥

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(*) 2018, 2020
Explain the quantum theory of Raman effect?

⇒ The occurrence of Raman scattering easily understood by quantum theory of radiation: The radiation of frequency ν of photon having energy $h\nu$, where k is plank's constant. photons undergo collisions with molecules & if the collision is perfectly elastic, they will be deflected unchanged. Elastic scattering is like a ball bearing bounches on the table without any loss of energy. since the energy is exchanged btwn photon & molecules during collision, then this collisions are called in-elastic. the molecule can gain or loss amounts of energy ΔE Joules. ΔE represent the change in the vibrational & rotational energy of the molecule. If the molecule gains energy ΔE, the photon will be scattered with energy $h\nu - \Delta E$ & the equivalent radiation will have a frequency $\nu - \frac{\Delta E}{h}$. If molecule loses energy ΔE then the energy of the scattered radiation $h\nu + \Delta E$ & the

frequency $\nu + \frac{\Delta E}{h}$.

2018, 2021, 2020, 2022

⊛ Difference btwn stokes & anti-stokes radiation?

⇒ If the radiation scattered with a lower frequency than that of the incident beam is called stokes radiation. It is accompanied by an increase in molecule energy. But If the radiation scattered with a higher frequency than that of the incident beam is called anti-stokes radiation. It is accompanied by a decrease in molecules energy. stokes radiation Generally more intense than anti-stokes. the intensity of stokes radiation is usually stronger than antistokes radiation because most molecules start from the ground state at typical temperature. stokes can be started from ground state but anti-stokes cannot started from ground state

2018, 2021, 2020, 2017, 2015

⊛ State the criteria for a molecule to be Raman active? Give 3 example which give Raman spectra?

⇒ A molecule is Raman active if it undergoes a change in its polarizability during vibration. Change in polarizability means the e⁻ cloud distribution needs to shift significantly

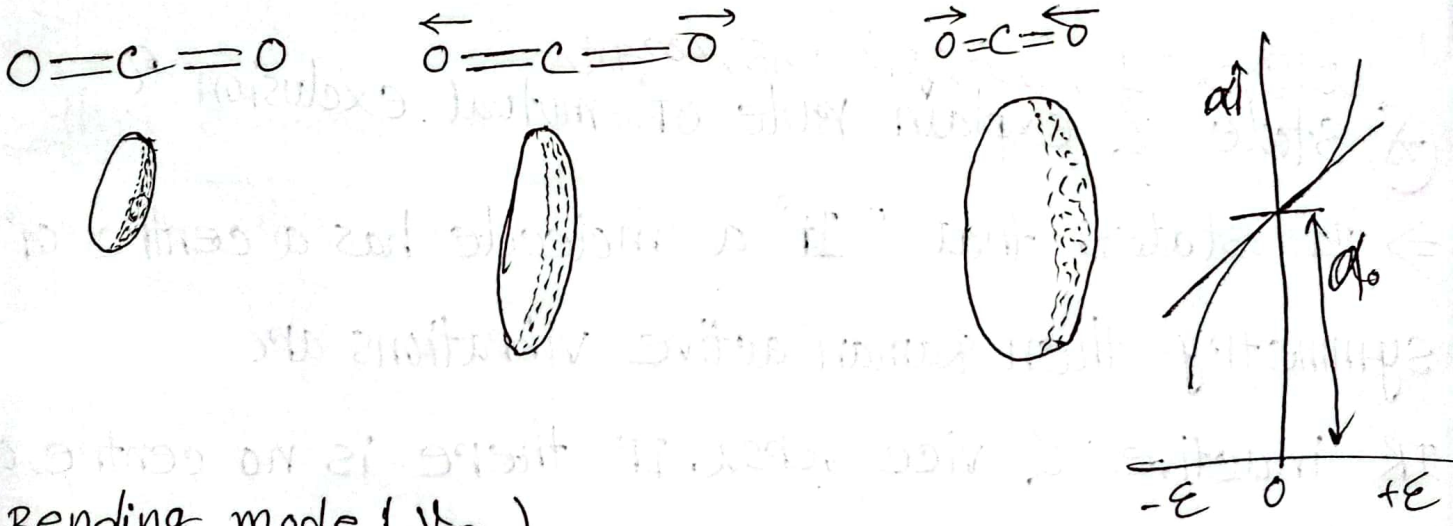
during the vibration. For a molecule to be Raman active it must have a vibrational mode that causes a change in its polarizability.

example: H_2O, C_6H_6, CO_2, N_2, Cl_2, CH_4, CCl_4, $CHCl_3$ etc.

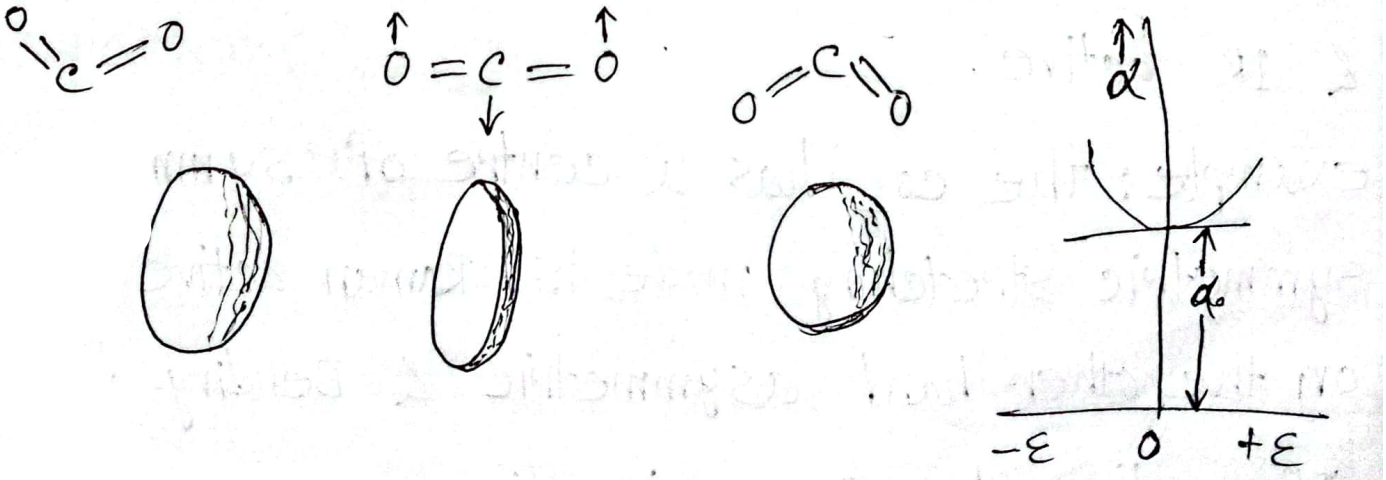
2018, 2015

(A) Symmetric stretching mode of CO_2 is Raman active, but bending & asymmetric modes is inactive. why?

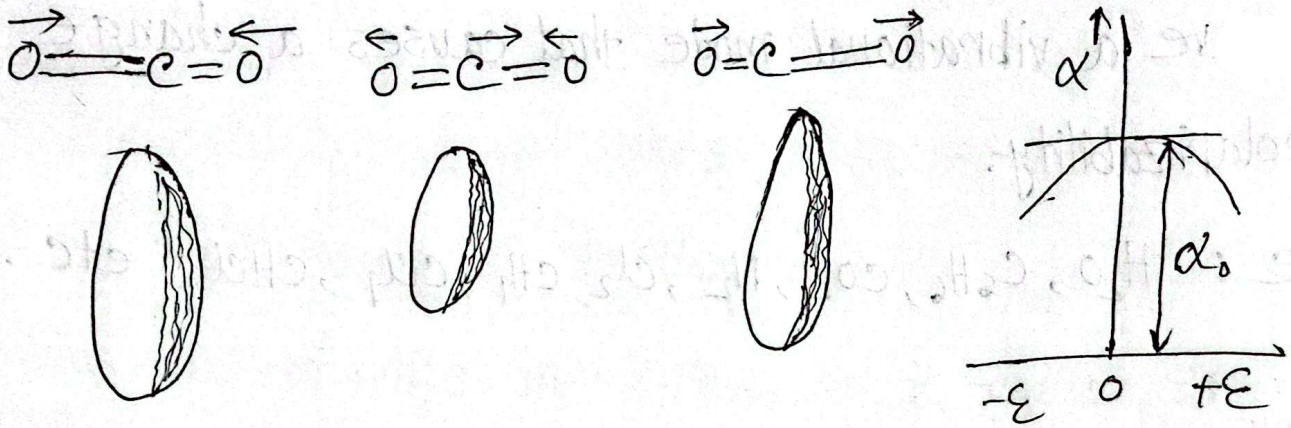
⇒ (I) Symmetric stretching mode (ν_1)



(II) Bending mode (ν_2)



(iii) asymmetric stretching mode:



But only in symmetric stretching mode change in polarizability so it is Raman active but other are not Raman active.

2018, 2021, 2020, 2017, 2015, 2024(H)

(*) State & explain rule of mutual exclusion?

⇒ It states that "If a molecule has a centre of symmetry then Raman active vibrations are ~~IR inactive~~ IR inactive & vice versa. If there is no centre of symmetry then some vibrations may be both Raman & IR active.

example: The CO_2 has a centre of symmetry. Its symmetric stretching mode is Raman active & IR inactive on the other hand asymmetric & Bending modes are IR active but Raman inactive. N_2O has no centre

of symmetry for this reason N_2O has common bands in IR & Raman spectra.

2018
(**) Explain why the pure rotations of spherical top molecules are completely inactive in Raman spectroscopy?

⇒ Examples of spherical top molecules are those with tetrahedral symmetry such as CH_4 or silane, SiH_4. The polarizability ellipsoid for such molecules is a spherical surface & it is evident that rotation of this ellipsoid will produce no change in polarizability. Therefore the pure rotations of spherical to molecules are completely inactive in the Raman.

2018, 2020
(**) State advantages of Raman over IR in vibrational measurements?

⇒ ① Firstly, because both the incident & scattered radiation are at UV or visible frequencies, conventional optics & sample cells (glass, quartz etc) can be used, so avoiding the problems inherent in NaCl windows,

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atmospheric absorption etc.

(ii) secondly, because the beam can be focused extremely finely, very small samples can be studied. This, combined with pulsed techniques which can give very short time resolutions, enables very small quantities of transient species to be studied.

(iii) thirdly, water, which has strong IR absorptions, is rather weak Raman ~~scha~~ scatterer and so aqueous soln can be studied using Raman, because the sample signal is not swamped by that of the solvent.

⊗ 2017, 2022 what is meant by Raman scattering?

⇒ If monochromatic radiation or radiation of a very narrow frequency band is used, the scattered energy will consist almost entirely of radiation of the incident frequency but in addition certain discrete frequencies

above & below that of the incident beam will be scattered; it is this which is Raman scattering.

⊛ 2017 write energy eqn for pure rotational Raman spectra for a linear molecule?

⇒ The rotational energy level for linear molecules,

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 \ cm^{-1} \quad (J = 0, 1, 2, \cdots)$$

where, B = Rotational constant

D = centrifugal distortion constant.

But, in Raman spectroscopy, ignoring D. then the simple expression for energy level,

$$\varepsilon_J = BJ(J+1) \ cm^{-1} \quad (J = 0, 1, 2, \ldots\ldots).$$

⊛ 2017 Show the rotational energy levels of a diatomic molecule & the rotational Raman spectrum arising from them with showing the relative intensities of lines in the spectrum?

⇒ The rotational energy levels of a linear diatomic

molecule:

$$\mathcal{E}_J = BJ(J+1) \quad cm^{-1}$$

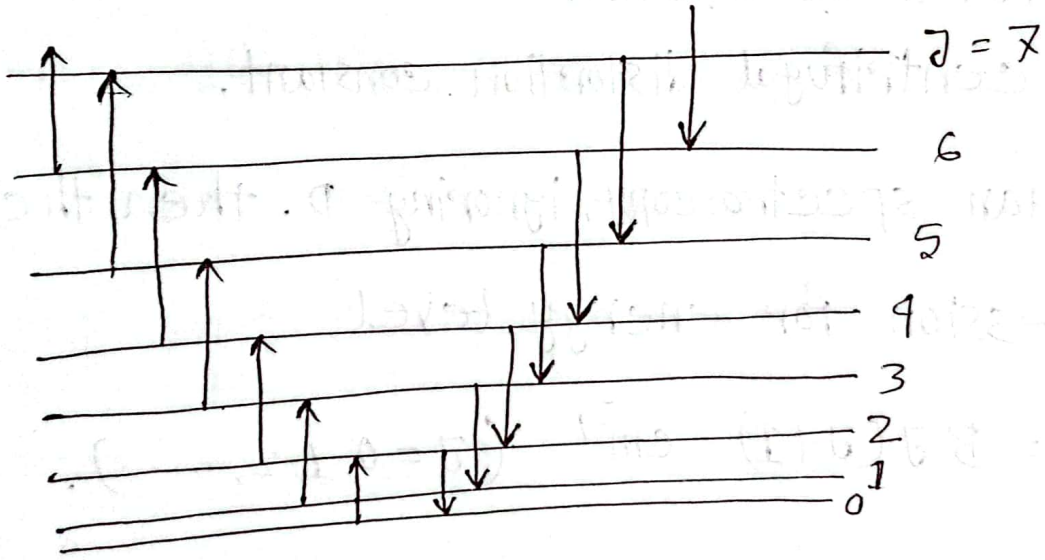
The selection rule $\Delta J = 0$ or ± 2 only.

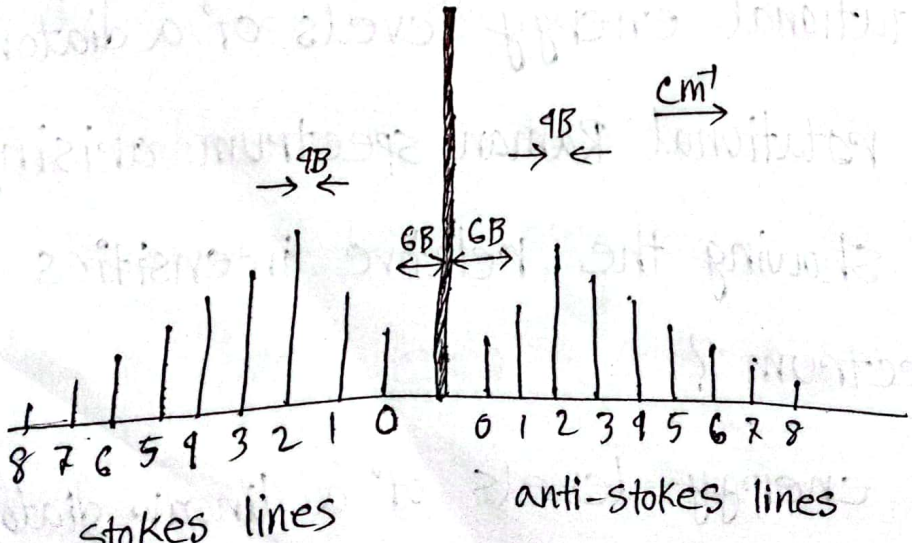
$\Delta J = 0$, corresponds to Rayleigh scattering, $\Delta J = \pm 2$ give

Rise to Raman line. when $\Delta J = \pm 2$

$$\Delta \mathcal{E} = \mathcal{E}_{J'=J+2} - \mathcal{E}_{J''=J}$$

$$= B(4J+6) \quad cm^{-1}$$





2017

Explain the stokes & anti-stokes lines in the Rotational Raman spectrum of the above fig-II ?

⇒ If the molecule gain rotational energy from photon during collision, there occurs a series of S-branch lines to the two wave number side then it is stokes line and if the molecule loses rotational energy to the photon during collision, then S-branch lines appear on the high wave number side (anti stokes line).

$$\bar{\nu}_S = \bar{\nu}_{ex.} \pm \Delta E_S$$

$$= \bar{\nu}_{ex.} \pm B(4J+6) \text{ cm}^{-1}$$

where plus sign refers to anti-stokes lines, the minus to stokes lines and $\bar{\nu}_{ex.}$ is the wavenumber of the exciting radiation.

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(★) 2017, 2015 what structural information can you get from Raman spectral analysis?

⇒ (i) Raman spectra can identify the presence of specific functional groups within a molecule, such as C=C, C≡C, C-H, C-O, & others. (ii) It provide information about the symmetry of a molecule that can helpful for distinguishing btwn different isomers. (iii) It can provide information about the conformation of polymer chains, such as whether they are crystalline or amorphous. (iv) It is used for measure the strain & stress within a material. (v) Raman can provide insights into molecular orientation in anisotropic material.

(★) 2015 write down the expression for pure raman rotational energy transition for a linear molecule?

⇒ The rotational energy levels of linear molecules,

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 \ cm^{-1} \quad (J = 0, 1, 2, \ldots)$$

In raman spectroscopy, the precision of the measurements does not normally warrant the retension of the term involving D, the centrifugal distortion constant. Thus we take simpler expression,

$$\varepsilon_J = BJ(J+1) \ cm^{-1} \quad (J = 0, 1, 2, \ldots)$$

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Selection rule : $\Delta J = 0$ or ± 2 only

for $\Delta J = +2$ the chang in energy levels,

$$\Delta E = E_{J'=J+2} - E_{J''=J} = B(4J+6) \text{ cm}^{-1}$$

As $\Delta J = +2$ so we may label these lines s branch lines & write,

$$\Delta E_s = B(4J+6) \text{ cm}^{-1} \quad (J = 0, 1, 2, \ldots)$$